Agri3D

3D Crop Structure Analysis from Field-Grown Maize Point Clouds

Semantic Segmentation, Full-Cloud Projection, Structural Reconstruction, and Phenotypic Geometry

Technical Report - Current Development Stage

Status: QUALIFIED WITH LIMITATIONS

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Guilherme Frutuoso de Almeida

Repository: github.com/GuilhermeFrAlmeida/agri3d-crop-phenotyping

Scope statement

This document records the current qualified stage of an ongoing research-engineering project. It is not the final project report. Additional model, robustness, external-data, and industrial-interface phases are planned.

Abstract

Agri3D is a reproducible 3D crop-structure analysis pipeline for field-grown maize point clouds. The current system combines deterministic point-cloud preprocessing, a frozen PointNet++ semantic segmentation baseline, nearest-neighbor projection from 10,000 sampled model points to 100,000-point source clouds, geometric leaf-component extraction, metric-space phenotype proxies, and checksummed artifact persistence. Experiments use 520 segmented MaizeField3D plants with a frozen plant-identity split of 364 train, 78 validation, and 78 untouched test plants. The best semantic model, an XYZ-only PointNet++ SSG configuration, achieved 0.8366 validation mIoU on the sampled 10k-point evaluation. Controlled augmentation, learning-rate scheduling, and estimated-normal feature experiments did not improve this baseline.

The frozen downstream pipeline was subsequently evaluated on all 78 validation plants in both oracle-semantic and model-predicted modes. All 156 plant-mode executions completed successfully with zero failures. Nearest-neighbor projection preserved semantic quality and slightly improved aggregate mIoU from 0.8366 on sampled points to 0.8397 on projected full clouds. Predicted semantics produced a mean of 17.92 structural components per plant versus 15.94 with oracle semantics, indicating a systematic tendency toward downstream fragmentation. Lower projected semantic mIoU was moderately associated with larger structural discrepancies (Spearman rho = -0.355, p = 0.0014). Instance reconstruction accounted for approximately 79% of predicted-mode end-to-end wall time, shifting the main engineering bottleneck from neural inference to CPU graph processing.

The current system is therefore qualified with limitations: semantic inference, full-cloud projection, deterministic structural extraction, metric-space measurements, and artifact generation are operational at validation scale, while biological leaf-instance accuracy and phenotype accuracy remain unqualified because verified instance and trait references are not yet available. The next project stages will evaluate stronger 3D neural architectures, robustness under controlled point-cloud perturbations, broader industrial interfaces, and cross-dataset or sensor-native validation including the Pullally LiDAR/GNSS dataset.

Keywords

3D computer vision; point clouds; maize phenotyping; PointNet++; semantic segmentation; LiDAR; structural reconstruction; digital agriculture; phenomics; geometric computing.

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1. Executive Summary

This report documents a current stage of Agri3D rather than a completed final system. The present milestone establishes a stable, model-neutral pipeline from annotated maize point clouds to semantic labels, full-resolution structural partitions, and auditable geometric phenotype proxies. The architecture was deliberately frozen before the 78-plant end-to-end evaluation so that the validation run served as evaluation rather than further tuning.

Table 1. Current qualification summary.

Area	Current evidence	Status
Semantic segmentation	Run 1 PointNet++ XYZ: 0.8366 validation mIoU on 78 plants	Frozen baseline
Full-cloud projection	0.8397 aggregate mIoU on 7.8M projected points; zero pipeline failures	Qualified
Instance structure	Deterministic graph components on 78 plants; oracle/predicted gap quantified	Qualified as structural extractor
Phenotype geometry	Height, component count, PCA/geodesic length proxies, hull/voxel volume, stalk descriptors	Computationally qualified
Biological validation	No verified leaf-instance GT or trait references available in current evidence	Not qualified
Interoperability	PyTorch .pt canonical; ONNX capture attempt failed before parity qualification	ONNX not qualified

The strongest system-level result is that the 10k-to-100k semantic projection does not introduce aggregate degradation. Sampled validation mIoU of 0.8366 became 0.8397 after full-cloud projection, with projected stalk IoU rising from 0.7530 to 0.7582. This makes the projection stage a credible bridge from the neural model to downstream geometry.

The principal current limitation is structural fragmentation under imperfect semantics. Across 78 plants, projected Run 1 semantics produced more accepted structural components than oracle semantics on 63 plants, the same count on 7, and fewer on 8. The mean component count increased from 15.94 to 17.92. Component counts are structural diagnostics and are not interpreted as verified biological leaf counts.

The next stage will not simply optimize mIoU in isolation. A stronger semantic architecture will be assessed against the frozen downstream system using semantic mIoU, full-cloud mIoU, structural consistency, phenotype stability, runtime, memory, and parameter count. Robustness, industrial interfaces, and external-data validation are separate planned phases.

2. Project Scope and Current Stage

2.1 Problem definition

The project asks a systems-level question: given a high-resolution 3D scan of a maize plant, can the plant be decomposed into useful structural classes and then converted into reproducible geometric measurements? The current task is intentionally narrower than complete biological phenotyping. It establishes semantic crop-structure segmentation, full-resolution reconstruction, deterministic structural components, and geometric proxies that can later be validated against biological references.

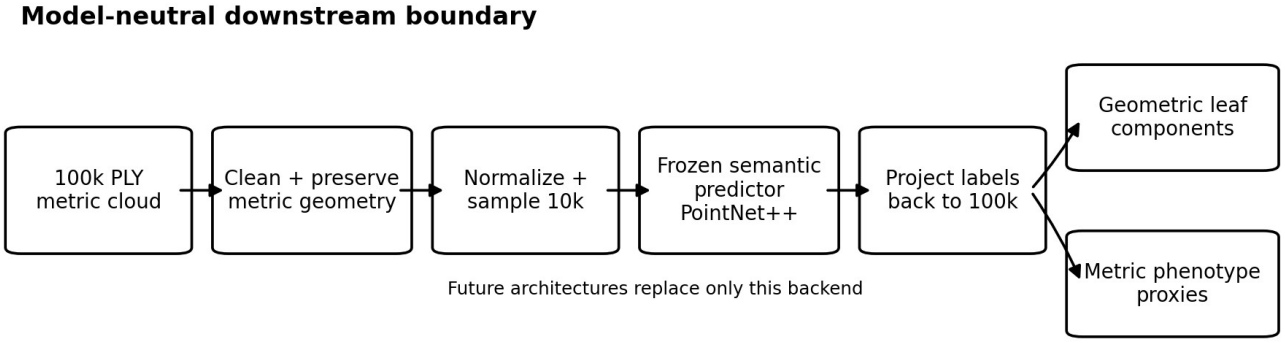


Figure 1. Current Agri3D inference-to-geometry architecture. The downstream boundary is model-neutral so that a future semantic backbone can replace PointNet++ without rewriting structural reconstruction or phenotype extraction.

2.2 Current-stage boundary

The current report ends after frozen validation of the PointNet++ semantic baseline and the Phase 4/5 downstream pipeline. The following are inside the scope of this report: deterministic preprocessing; semantic segmentation development; training-performance profiling; full-cloud label projection; geometric component extraction; metric-space phenotype proxies; runtime analysis; artifact provenance; and descriptive validation-scale qualification.

The following remain future work: a stronger 3D deep-learning backbone; controlled robustness experiments; verified biological instance and phenotype references; cross-dataset validation; sensor-native LiDAR integration; and a broader industrial interface including stable APIs, reporting services, deployment packaging, and operational qualification.

3. Dataset and Experimental Contract

3.1 MaizeField3D

Agri3D uses the segmented 100k-point subset of MaizeField3D, a curated dataset of field-grown maize point clouds. The associated dataset publication reports 1,045 total point clouds and 520 segmented/annotated plants, with multi-resolution point-cloud products and metadata. Agri3D uses the 520 annotated plants for supervised development [1].

Within the segmented PLY export used here, XYZ provides geometry and RGB provides annotation. The project decodes black RGB points as stalk and non-black RGB points as leaf. RGB is never used as a neural input feature because it directly defines the semantic target in this dataset representation.

Table 2. Frozen experimental contract.

Contract item	Frozen value
Annotated plants	520
Source cloud	100,000 points per plant
Model sample	10,000 deterministic points per plant
Classes	0 = leaf; 1 = stalk
Train / validation / test	364 / 78 / 78 plants
Split unit	Plant identity

Contract item	Frozen value
Seed	42
Primary model-selection metric	Validation mIoU
Test usage	Untouched in all reported experiments
Neural loss	Masked, unweighted cross-entropy
Optimizer	Adam, initial learning rate 0.001

3.2 Evaluation levels

Two evaluation levels are intentionally separated. Neural experiments report metrics on the fixed 10,000-point samples used by the model. The downstream pipeline then projects those predictions to the cleaned 100,000-point source cloud and reports a separate full-cloud semantic evaluation. The original 10k result is not replaced by the projected result; both are retained because they answer different questions.

The 78 validation plants have informed model selection and downstream qualification, so they are not interpreted as an independent final test estimate. The frozen 78-plant test partition remains unopened and is outside the present report.

3.3 Evidence base

The report was assembled from frozen experiment artifacts rather than manually transcribed notebook screenshots. The evidence base includes semantic validation JSON files for the geometry baseline and Runs 1-4, the Run 1 per-plant validation analysis, Phase 3P profiling outputs, the bounded Phase 4/5 val5 artifacts, the Phase 4A frozen configuration report, the complete val78 aggregate bundle, and a complete frozen-config per-plant artifact bundle for plant 0495. This supports both aggregate statistics and qualitative sanity checks while preserving experiment provenance.

4. System Architecture and Methods

4.1 Preprocessing and semantic prediction

The model pipeline reads PLY geometry, removes non-finite XYZ while preserving aligned source indices, retains the original metric-space cloud for downstream geometry, normalizes the clean cloud for neural inference, and deterministically samples 10,000 points. PointNet++ receives XYZ only in the frozen Run 1 baseline. Model output is a per-point binary logit tensor for leaf and stalk.

4.2 Model-neutral semantic boundary

A key engineering change in Phases 4/5 was to isolate semantic inference behind a typed SemanticPredictor interface. Downstream projection, instance extraction, phenotype measurements, and artifact generation do not import PointNet++ internals. This separation is important because the next neural architecture can be evaluated as a replacement semantic backend rather than as a rewrite of the complete system.

4.3 Full-cloud projection

The frozen projection method builds a metric-space nearest-neighbor index over valid sampled points and assigns every cleaned source point the label and probability of its nearest sampled point. Predictions for sampled source rows are explicitly restored by source identity. Exact-distance ties are resolved deterministically. No learned upsampling, voting, or post-processing is used in the current baseline.

4.4 Structural reconstruction

Leaf points are converted into a capped radius graph. Connected components form initial leaf-structure candidates; components below a minimum size are rejected as unassigned noise. Instance IDs are

deterministic. Stalk points retain a dedicated structural label. The frozen configuration uses a 0.020 m neighborhood radius, at most 32 neighbors, a 100-point minimum accepted component size, and a 0.03 m attachment-diagnostic radius.

4.5 Phenotype proxies

All physical measurements operate on the cleaned metric-space cloud, not on unit-sphere-normalized model coordinates. Height is the full-cloud vertical extent. Leaf count in the current schema is the number of accepted geometric components and therefore remains a structural diagnostic rather than verified biological leaf count. Leaf length is represented by a PCA-axis extent and an optional bounded graph-geodesic proxy. Canopy volume is represented separately by convex-hull volume and voxel occupancy; the two methods are not treated as interchangeable physical quantities. Stalk descriptors include point count, centroid, vertical extent, and a canonical-sign PCA axis.

5. Semantic Segmentation Development

5.1 Controlled experiment sequence

PointNet++ [2] was selected as the first learned point-cloud baseline. The experimental sequence deliberately changed one major factor at a time while holding the frozen split, sampling policy, architecture family, optimizer, batch size, and validation selection metric fixed. This produced a transparent record of both successful and unsuccessful interventions.

Table 3. Semantic segmentation experiment registry.

Experiment	Controlled change	Best epoch	mIoU	Leaf IoU	Stalk IoU	Decision
Geometry baseline	Classical geometry	-	0.6230	0.8173	0.4286	Reference
Run 1	PointNet++ XYZ	76	0.8366	0.9202	0.7530	Frozen baseline
Run 2	Z-rotation + scale + jitter	93	0.8296	0.9199	0.7393	Rejected
Run 3	ReduceLROnPlateau	99	0.8317	0.9186	0.7449	Rejected
Run 4	XYZ + estimated normals	76	0.8173	0.9085	0.7261	Rejected

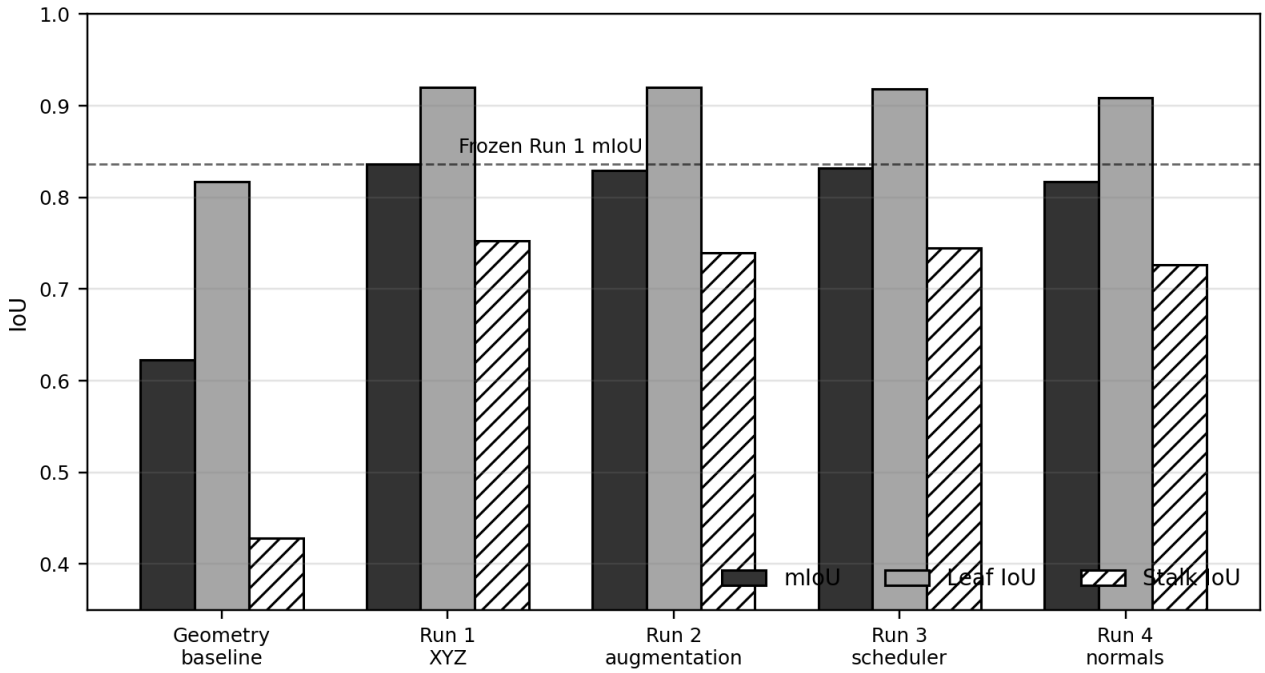


Figure 2. Validation IoU comparison across the classical geometry baseline and four PointNet++ experiments. Run 1 remained strongest after augmentation, scheduler, and estimated-normal interventions.

5.2 Interpretation

Run 1 established the current semantic ceiling within the tested PointNet++ configuration. Run 2 increased stalk precision to 0.8828 but reduced stalk recall to 0.8197 and lowered aggregate mIoU to 0.8296. Run 3 changed optimization behavior through validation-driven learning-rate reduction but reached only 0.8317 mIoU. Run 4 added deterministic kNN-PCA surface normals as point features and fell to 0.8173 mIoU; stalk precision decreased to 0.8194 while recall remained near the Run 1 level. These results support an architecture-level change rather than continued small modifications to the same baseline.

The negative experiments are retained because they narrow the design space. They do not prove that augmentation, scheduling, or normals are universally harmful; they show that the evaluated policies did not improve the frozen development contract.

6. Frozen Semantic Baseline

6.1 PointNet++ SSG configuration

The frozen semantic backend is a pure-PyTorch PointNet++ single-scale grouping implementation. Set abstraction reduces 10,000 input points to 1,024, 256, 64, and 16 centroids using radii 0.1, 0.2, 0.4, and 0.8 with 32 neighbors at each stage. Feature propagation restores features to all input points and a per-point head produces two logits. The selected Run 1 checkpoint is epoch 76 with SHA256 608a8f15319b0fc17daeefed92f1c278fd638a82adf30768c1c6d05a21ef03a9.

Table 4. Frozen Run 1 semantic performance.

Metric	Run 1 validation
Accuracy	0.9358
mIoU	0.8366
Leaf IoU	0.9202
Stalk IoU	0.7530
Stalk precision	0.8521

Metric	Run 1 validation
Stalk recall	0.8662
Stalk F1	0.8591
Evaluated points	780,000
Validation plants	78

6.2 Validation distribution and difficult cases

Per-plant sampled-point mIoU had mean 0.8395, median 0.8450, standard deviation 0.0669, minimum 0.6171, and maximum 0.9568. The weakest plant was 0022, while the strongest listed case was 0408. The difficulty is concentrated more strongly in stalk segmentation than leaf segmentation, consistent with the class imbalance and thinner, more ambiguous stalk geometry.

Plant 0022 is also the only validation plant in the metadata analysis marked with touching leaves, but n = 1 is insufficient to infer a general effect. The report therefore treats this as a qualitative stress case rather than a statistical condition result.

6.3 Training-performance engineering

Phase 3P profiling on a Tesla T4 measured approximately 2.58 samples/s at the scientific batch size of 2, with peak allocated GPU memory of 310 MiB. In the detailed batch profile, forward computation represented 45.2% of measured batch time, data wait 43.4%, and backward computation 10.2%. The first set-abstraction stage was the dominant geometry hotspot. The low memory footprint and geometry-heavy operator profile indicate that optimized point-cloud kernels, rather than additional VRAM, are the more relevant future training optimization target.

7. Full-Cloud Semantic Projection

7.1 Frozen val78 result

Table 5. Sampled and full-cloud semantic performance on the frozen 78-plant validation run.

Metric	Sampled 10k	Projected 100k	Aggregate delta
Accuracy	0.93585	0.93676	+0.00091
mIoU	0.83662	0.83967	+0.00305
Leaf IoU	0.92025	0.92111	+0.00086
Stalk IoU	0.75299	0.75822	+0.00523
Stalk precision	0.85214	0.85564	+0.00350
Stalk recall	0.86615	0.86944	+0.00328

Aggregate mIoU increased by +0.0030. At the plant level, the mean projected-minus-sampled mIoU delta was +0.0026 and the median was +0.0016; 49 plants improved and 29 decreased. The most negative observed plant-level delta was -0.0195, while the largest positive delta was +0.0294. These results support nearest-neighbor projection as a practical baseline rather than evidence that projection universally improves individual plants.

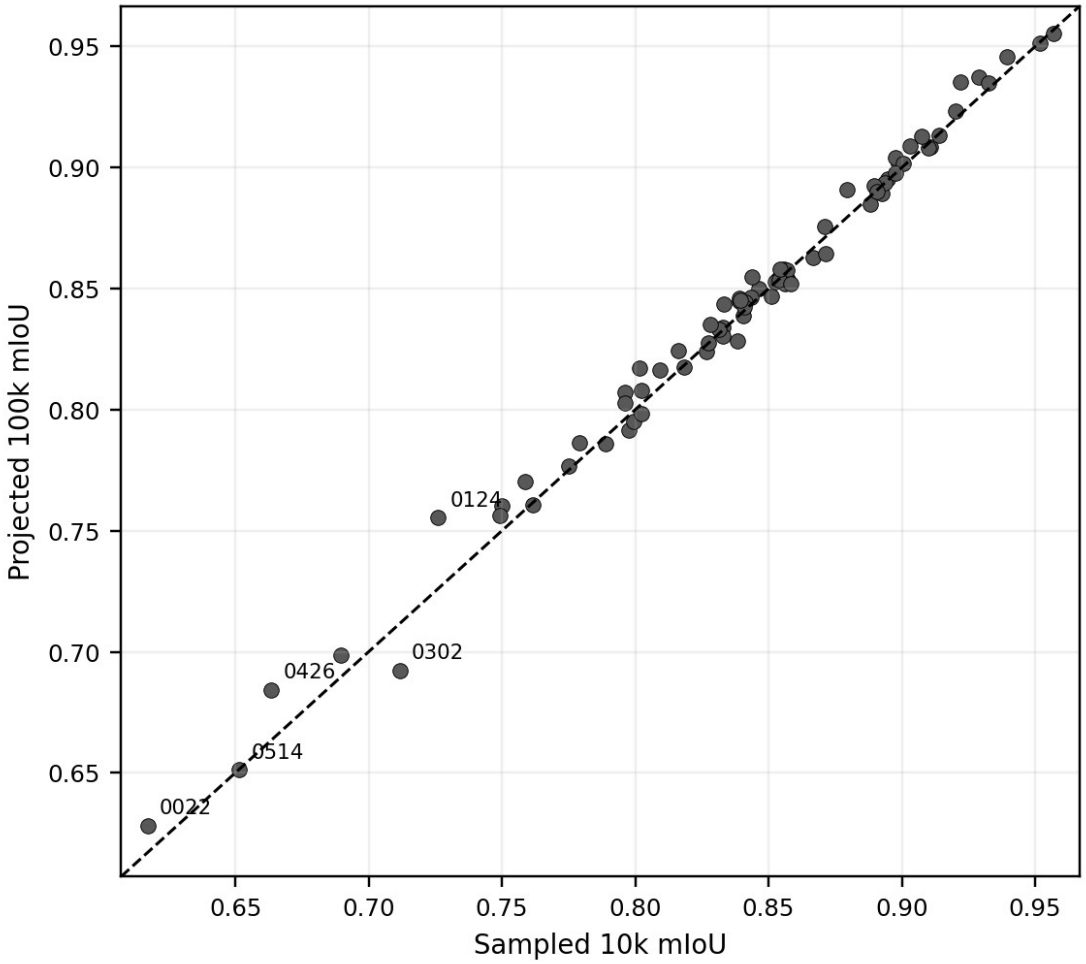


Figure 3. Per-plant sampled 10k mIoU versus projected 100k mIoU. The dashed line denotes equal performance.

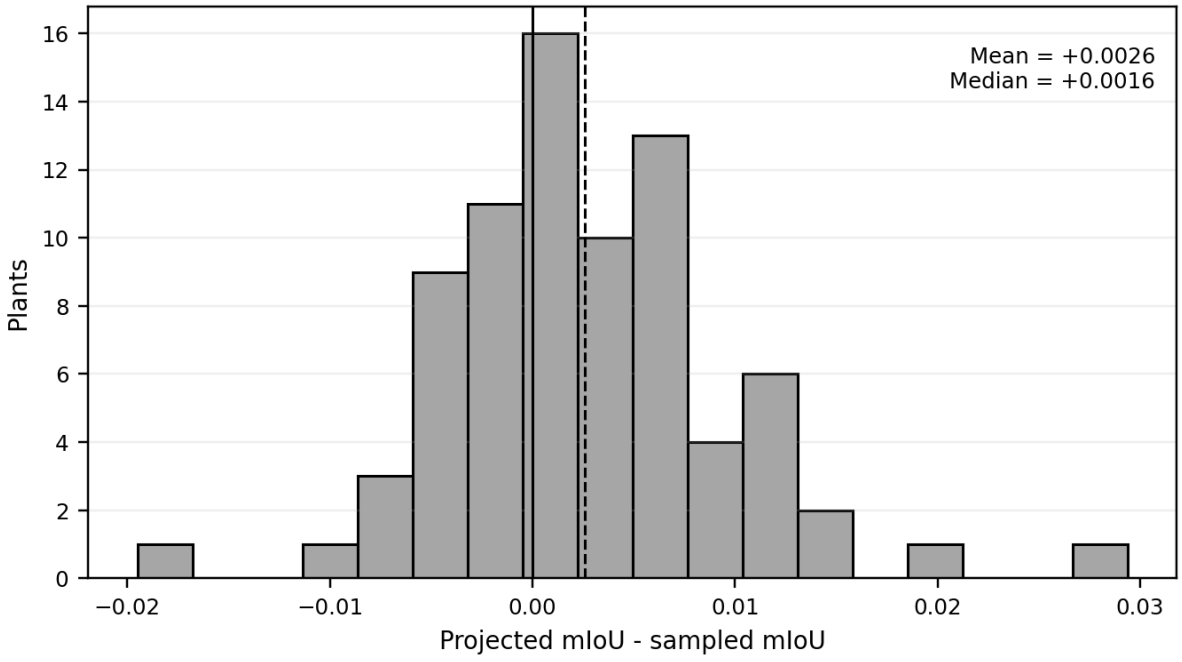


Figure 4. Distribution of per-plant projection deltas. Aggregate and median behavior is close to neutral with a small positive shift, while individual plants can move in either direction.

7.2 Qualification interpretation

Because projection preserves aggregate semantic quality on 7.8 million evaluated full-cloud points and does not introduce pipeline failures, it is qualified as the current full-cloud reconstruction method. This conclusion is limited to the present data contract, density, deterministic sampling, and nearest-neighbor geometry. It does not establish optimality relative to learned upsampling or other interpolation schemes.

8. Qualitative Semantic Analysis

Quantitative IoU alone does not reveal where crop-structure errors occur. Standardized point-cloud views were therefore used to compare ground-truth semantic labels, projected predictions, and error locations. Plant 0006 represents a comparatively strong case, while 0022 is a known difficult case with substantially lower semantic mIoU and touching-leaf metadata.

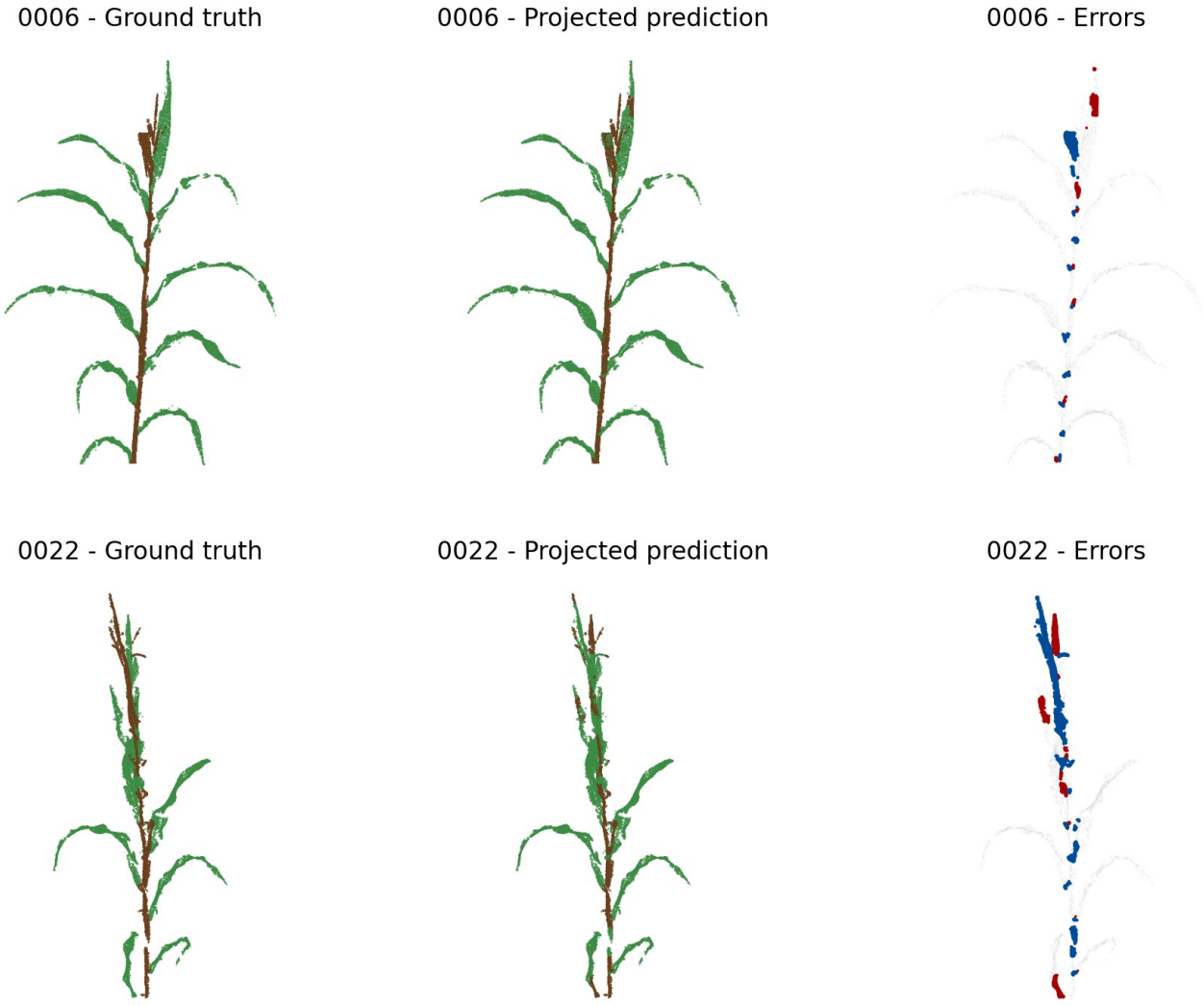


Figure 5. Qualitative semantic comparison for plants 0006 and 0022. Errors cluster around stalk/leaf boundaries and upper structural regions; 0022 shows substantially broader stalk-to-leaf and leaf-to-stalk confusion.

The qualitative evidence supports two engineering conclusions. First, large portions of leaf geometry remain stable even on difficult plants, which explains why aggregate leaf IoU is high. Second, errors concentrated along stalk and attachment regions can materially alter graph connectivity downstream even

when overall pointwise accuracy remains high. This motivates evaluating future neural architectures not only by mIoU but also by downstream structural consistency.

9. Instance Reconstruction and Phase 4A Configuration Study

9.1 Structural baseline

The structural extractor is intentionally conservative and geometry-first. It does not attempt neural instance segmentation. Leaf points are linked through a metric radius graph, connected components are computed, components below 100 points are rejected, and deterministic positive instance IDs are assigned to accepted components. The initial development setting used a 0.015 m radius and 16 neighbors.

9.2 Bounded Phase 4A sweep

Before full validation, a bounded five-plant study evaluated 12 graph configurations spanning radii of 0.010, 0.015, 0.020, and 0.025 m and neighbor caps of 16, 24, and 32. Minimum component size and attachment radius were held fixed. The five-plant cohort was used as a development/qualification subset and was not treated as an independent final validation cohort.

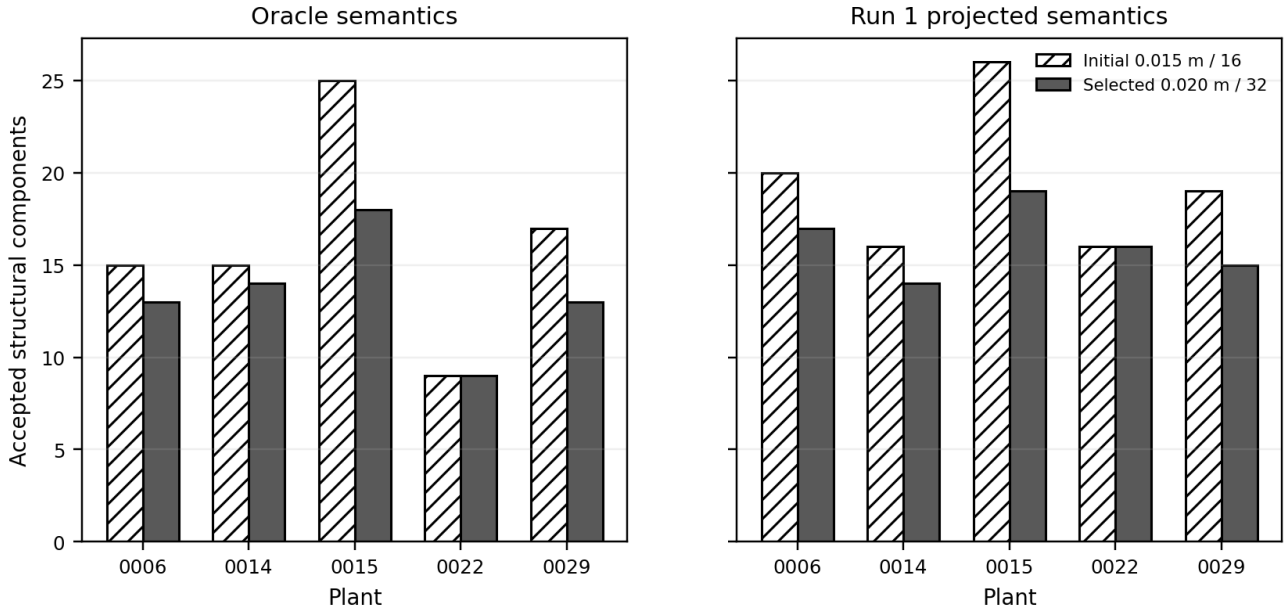


Figure 6. Accepted structural components under the initial and selected graph configurations on the five-plant development cohort. Lower counts were interpreted together with visual review and fragmentation diagnostics, not as biological leaf-count accuracy.

Table 6. Phase 4A component-count diagnostics.

Plant	Oracle: initial → selected	Predicted: initial → selected
0006	15 → 13	20 → 17
0014	15 → 14	16 → 14
0015	25 → 18	26 → 19
0022	9 → 9	16 → 16
0029	17 → 13	19 → 15

The selected setting, 0.020 m / 32 neighbors, reduced oracle near-threshold accepted components from 9 of 81 to 4 of 67 while keeping candidate unassigned fractions below 0.177% across oracle and predicted modes. CPU extractor mean time increased from 3.22 s to 4.14 s. The selection therefore represents a documented tradeoff: greater connectivity and lower small-fragment behavior at higher CPU cost.

The experiment explicitly did not claim biological leaf recovery. Plant 0015 remained parameter-sensitive and plant 0022 retained a seven-component oracle/predicted gap, indicating that graph connectivity alone cannot resolve errors originating from semantic prediction.

10. Frozen 78-Plant End-to-End Evaluation

10.1 Execution integrity

The frozen configuration was executed once across all 78 validation plants in both oracle-semantic and Run 1 projected-semantic modes. The run produced 156 successful plant-mode executions and 0 failures. No parameter tuning was performed after observing the full validation results.

10.2 Structural component behavior

Table 7. Structural diagnostics across the frozen validation cohort.

Statistic	Oracle semantics	Projected Run 1 semantics
Mean accepted components	15.94	17.92
Median	15	17
Minimum	8	10
Maximum	28	29
Mean unassigned leaf fraction	0.145%	0.270%
Maximum unassigned leaf fraction	0.594%	0.953%

Projected semantics produced more accepted components than oracle semantics on 63 of 78 plants, the same count on 7, and fewer on 8. The mean signed difference was +1.99 components per plant, with median absolute difference 2 and a maximum absolute difference of 7. This is consistent with a tendency toward fragmentation under semantic error, but it is not a biological leaf-count error measurement.

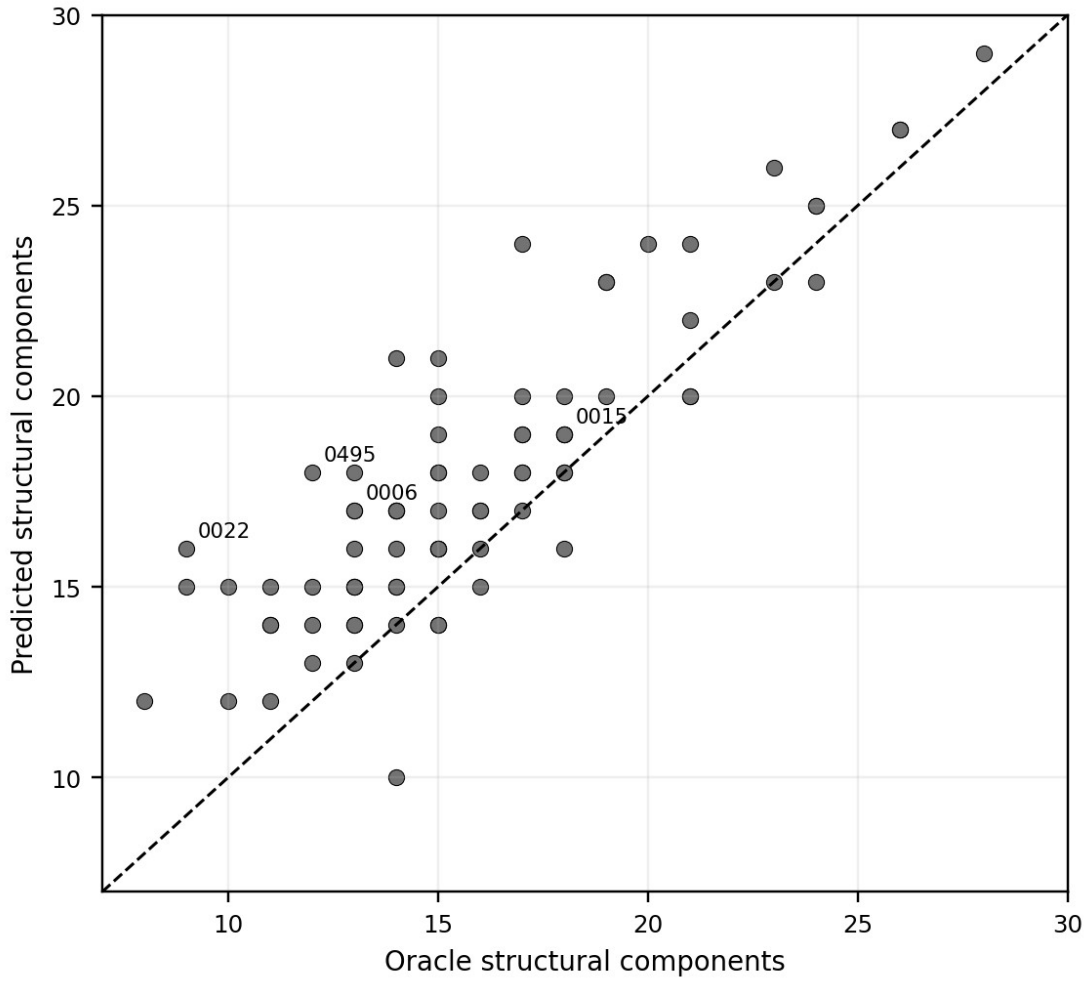


Figure 7. Oracle versus predicted structural component counts for all 78 validation plants. Points above the diagonal indicate more components under projected model semantics.

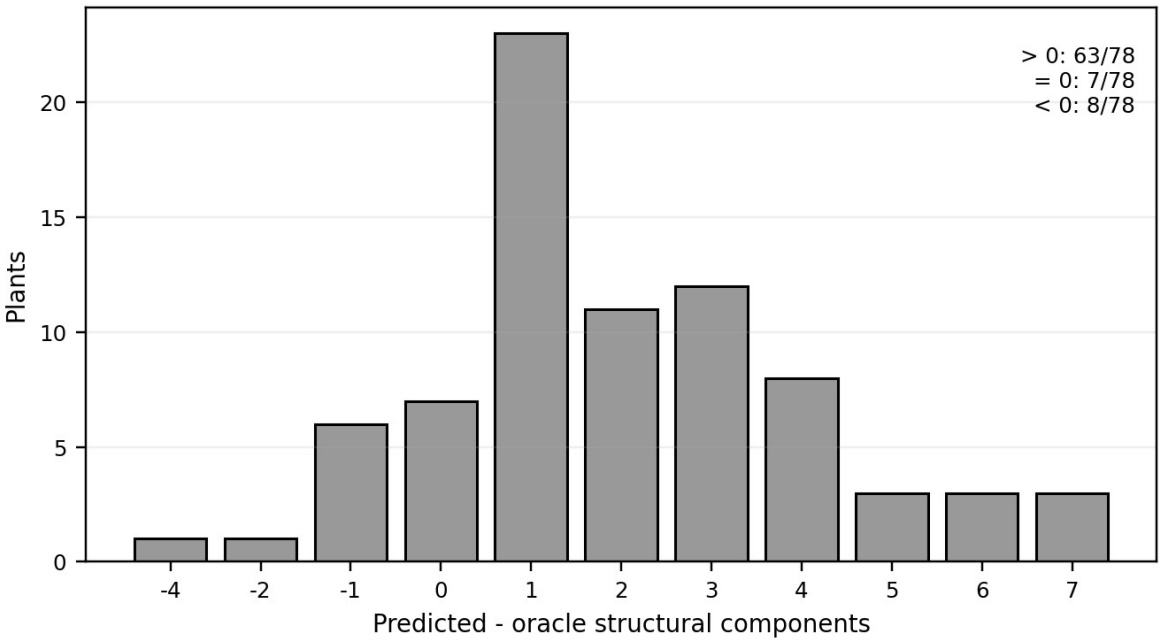


Figure 8. Distribution of the signed structural component difference between predicted and oracle semantic modes.

10.3 Semantic quality and structural discrepancy

Projected semantic quality was moderately associated with structural disagreement: Spearman rho = -0.355 (p = 0.0014) between projected full-cloud mIoU and absolute oracle/predicted component-count difference. Pearson r was -0.449. This is descriptive evidence, not proof of causality, but it supports the hypothesis that semantic errors propagate into the structural partition.

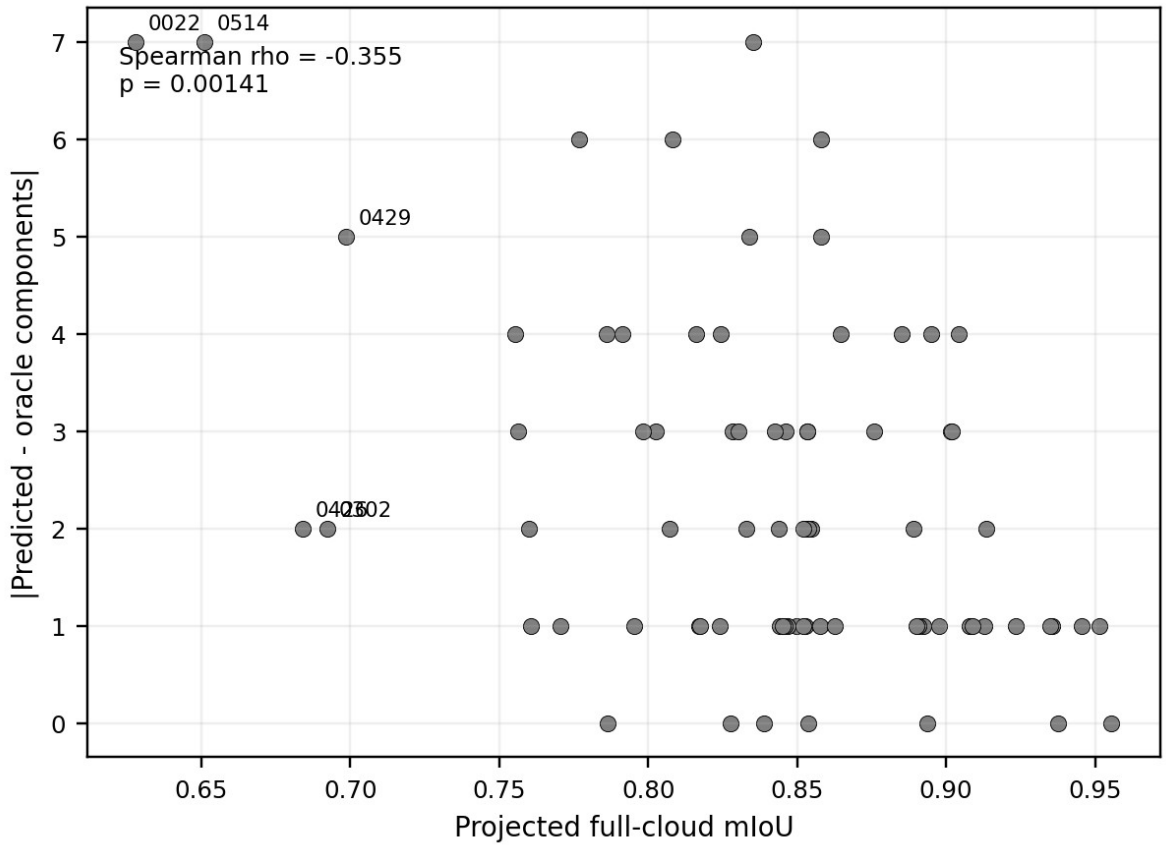


Figure 9. Projected full-cloud mIoU versus absolute structural component discrepancy. Lower semantic quality tends to coincide with larger downstream gaps, although substantial plant-specific variation remains.

10.4 Frozen-config qualitative example

Plant 0495 provides a complete frozen-config artifact example. Oracle semantics produced 12 accepted components while projected Run 1 semantics produced 18. Both modes preserved identical plant height because height is computed from full metric geometry independently of semantic labels. The predicted mode classified more points as leaf, increasing graph complexity and component count.

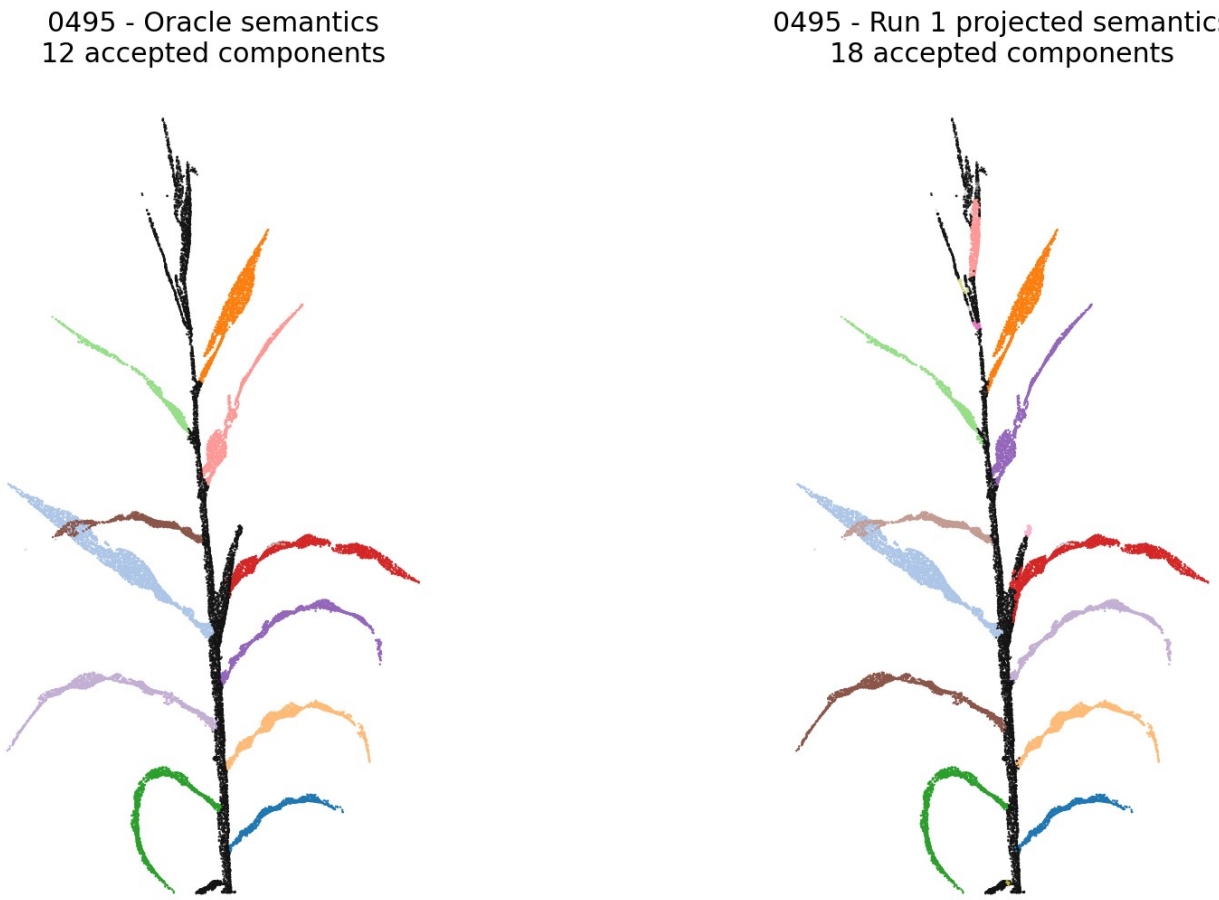


Figure 10. Frozen-config structural extraction for plant 0495. Colors denote geometric components; black denotes stalk. The predicted semantic mode produces additional fragmented components relative to the oracle mode.

11. Phenotypic Geometry Outputs

11.1 Plant height

Height is computed directly from the cleaned full-cloud Z extent and is therefore invariant to semantic mode. Across the 78 validation plants, mean height was 1.848 m, median 1.864 m, minimum 1.142 m, and maximum 2.768 m. Oracle and predicted modes were identical for every plant by design.

11.2 Structural count

The current leaf_count field is the number of accepted geometric leaf components. It is useful as an internal structural diagnostic and for comparing oracle and predicted modes, but it must not be interpreted as verified botanical leaf count until a reviewed instance palette or independent reference is available.

11.3 Leaf-length proxies

The pipeline reports a straight PCA-axis extent and a bounded graph-geodesic proxy. Across the 78 plants, the mean per-plant geodesic-minus-PCA difference was 0.080 m in oracle mode and 0.072 m in predicted mode. The systematically positive difference is geometrically plausible for curved leaves, but neither method is treated as biologically validated midrib length.

11.4 Canopy-volume proxies

Convex-hull and voxel-occupancy volumes intentionally measure different geometric concepts. Mean convex-hull volume was 0.295 m^3 in oracle mode and 0.302 m^3 in predicted mode. Mean voxel occupancy

was 0.00525 m³ and 0.00525 m³, respectively. The much smaller voxel value reflects occupied point-surface cells rather than a filled plant envelope; the methods should therefore be retained as separately named proxies rather than collapsed into one ambiguous volume metric.

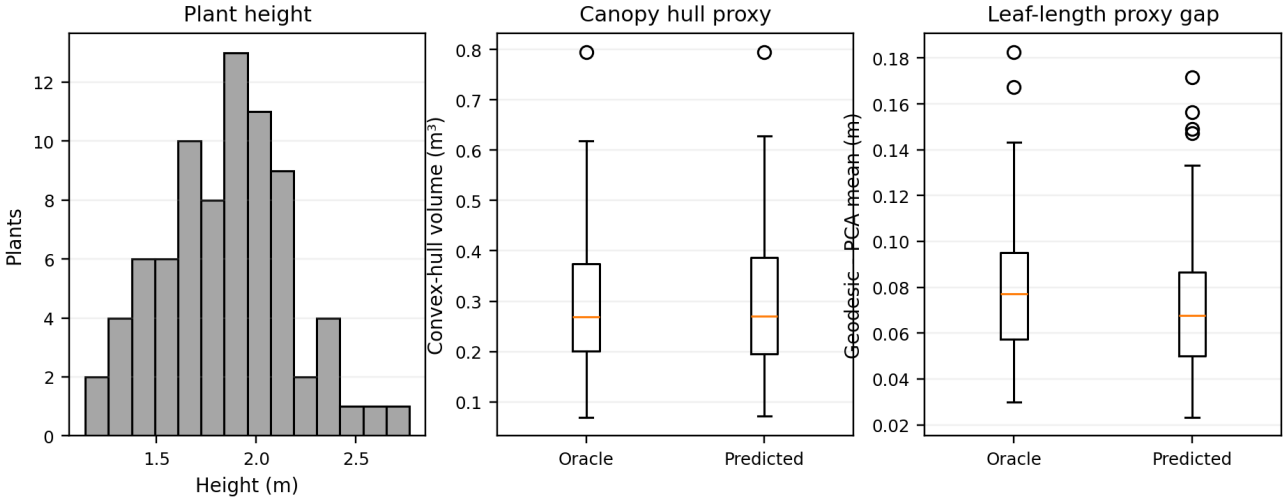


Figure 11. Validation distributions for plant height, convex-hull volume, and the geodesic-minus-PCA leaf-length proxy difference.

12. Runtime and Systems Engineering

12.1 End-to-end val78 timing

Table 8. Predicted-mode wall-time profile across 78 validation plants.

Stage	Mean (s)	Median (s)	P95 (s)	Max (s)
Semantic inference	0.785	0.691	1.059	1.550
Full-cloud projection	0.273	0.238	0.397	0.435
Instance extraction	7.670	7.476	9.253	16.429
Phenotype extraction	0.563	0.515	0.823	0.892
Sampling	0.194	0.159	0.304	0.576
Serialization	0.071	0.072	0.090	0.128
Total	9.693	9.484	11.025	19.255

Instance extraction is now the dominant end-to-end bottleneck, accounting for approximately 79.1% of mean predicted-mode wall time. Mean semantic inference is 0.785 s and projection 0.273 s, while instance extraction averages 7.670 s. This shifts the main optimization target from GPU inference to CPU graph construction and connected-component processing.

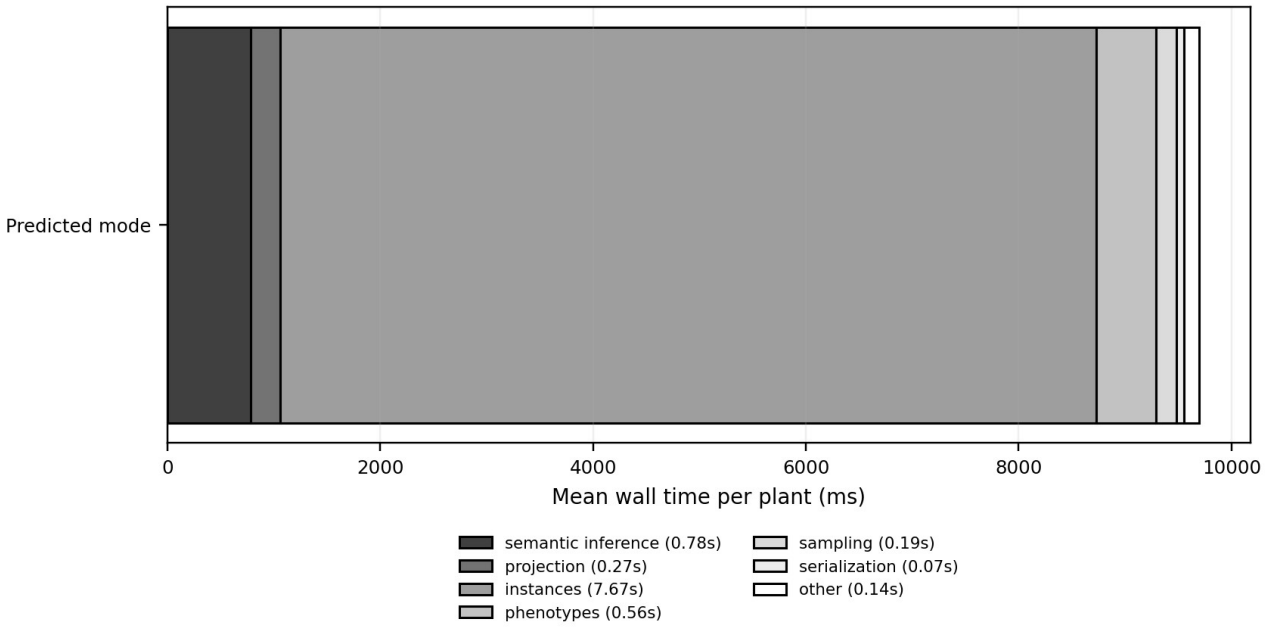


Figure 12. Mean predicted-mode wall-time composition. The geometric instance stage dominates the current end-to-end pipeline.

12.2 Training-side performance observations

Training profiling reached the complementary conclusion that the PointNet++ implementation is geometry-operator limited rather than memory limited. On a Tesla T4, peak allocated memory remained near 325 MiB at batch size 2. Disposable throughput tests increased from approximately 2.58 samples/s at batch size 2 to 3.58 at batch size 4 and 4.28 at batch size 8, but the official scientific experiments retained batch size 2 because changing batch size alters optimizer grouping and BatchNorm statistics.

Together, the training and inference profiles suggest two distinct optimization tracks: optimized point-cloud kernels for neural training/inference and a more efficient CPU or compiled graph implementation for downstream instance reconstruction. Neither optimization is required to interpret the scientific results reported here.

13. Qualification Status and Limitations

Table 9. Qualification matrix for the current project stage.

Subsystem	Qualification statement
Run 1 semantic model	Qualified on the frozen validation split; test remains untouched.
10k → 100k projection	Qualified for the current pipeline contract; aggregate semantic quality preserved.
Instance extraction	Qualified as deterministic structural extraction, not as biological leaf-instance accuracy.
Plant height	Computationally qualified in metric space; biological reference error not evaluated.
Structural leaf count	Diagnostic only; not verified botanical leaf count.
Leaf length	PCA and graph-geodesic geometric proxies; not biologically validated.
Canopy volume	Method-specific geometric proxies; not biological volume ground truth.
Artifact/provenance pipeline	Qualified at validation scale; 156/156 plant-mode executions completed.
ONNX backend	Not qualified; capture failed before checker/runtime/parity qualification.
External generalization	Not yet evaluated.

13.1 Primary limitations

- All neural model comparisons use a single seed and therefore do not establish statistical significance across training seeds.
- The reported semantic model was selected using the validation split; the untouched test split is reserved for a later final qualification stage.
- The split is grouped by plant identity, not genotype-held-out.
- No independent verified leaf-instance ground truth was available in the current pipeline evidence, so component counts cannot be interpreted as botanical leaf-count accuracy.
- No verified biological references for plant height, leaf length, or canopy volume were used in this stage; current outputs are reproducible geometric measurements or proxies.
- Nearest-neighbor projection is qualified only for the current data density and preprocessing contract; it has not been compared against learned projection methods.
- PointNet++ remains a pure-PyTorch implementation with known geometry-operator performance limits.
- Instance reconstruction is CPU dominated and is the primary current end-to-end latency bottleneck.
- Cross-dataset and sensor-native LiDAR generalization remain untested.

14. Discussion

14.1 What the current stage establishes

The project has moved beyond a standalone segmentation experiment. It now has a stable chain from source PLY through sampled neural inference, full-cloud semantic reconstruction, structural partitioning, geometric measurements, and checksummed outputs. This matters because future model improvements can be evaluated by their impact on the entire system rather than only on a pointwise segmentation score.

14.2 Architecture ceiling within the tested PointNet++ family

Run 1 remained strongest after three controlled interventions. Augmentation, validation-driven scheduling, and explicit estimated normals all failed to improve mIoU under the frozen contract. These experiments do not exhaust the PointNet++ design space, but additional small hyperparameter modifications now have lower expected information value than a controlled comparison against a stronger point-cloud architecture.

14.3 Projection is not the current bottleneck

The full-cloud projection step preserved aggregate semantic quality and is computationally inexpensive relative to instance reconstruction. The evidence therefore does not justify replacing projection before addressing the semantic backbone or downstream graph efficiency. Individual plants can still improve or degrade after projection, so per-plant behavior should remain part of future robustness analysis.

14.4 Semantic errors propagate into structure

The oracle-versus-predicted structural comparison provides a useful systems metric. Predicted semantics increased accepted component count on most plants, and lower full-cloud mIoU was moderately associated with a larger structural discrepancy. This demonstrates why future semantic models should be assessed not only by mIoU but also by downstream structural stability. A model that gains a small amount of mIoU but substantially reduces fragmentation could be more useful for phenotyping than a model whose pointwise gain does not improve organ-level geometry.

14.5 Engineering bottleneck migration

At earlier stages, GPU training and pure-PyTorch point-cloud operations were the obvious performance concerns. After freezing the current inference pipeline, the dominant wall-time cost is the CPU structural graph. This is a normal system-evolution outcome: once one stage is stabilized, the bottleneck can move downstream. The current evidence therefore provides a concrete optimization target for later industrialization work.

15. Next Development Stages

This report marks a current milestone, not project completion. The planned continuation is intentionally broader than simply replacing PointNet++. The next stages are organized around model quality, robustness, external validity, and industrial usability.

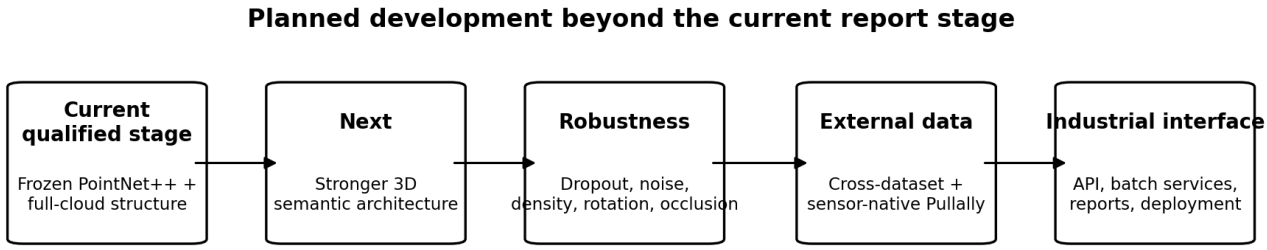


Figure 13. Planned development beyond the current report stage. The sequence is conceptual rather than a fixed calendar.

15.1 Stronger 3D semantic architecture

The next neural experiment will evaluate a stronger point-cloud architecture while preserving the frozen downstream interface. Candidate families include Point Transformer or KPConv-style models. The new model should be compared against Run 1 under the same split, 10k sampling contract, validation metrics, full-cloud projection, and downstream evaluation. Model comparison should include semantic mIoU, class IoU, full-cloud mIoU, structural component stability, phenotype-proxy stability, runtime, GPU memory, and parameter count.

15.2 Robustness experiments

A dedicated robustness phase will introduce controlled perturbations rather than relying only on clean curated PLY inputs. Planned perturbations include point dropout, reduced point density, Gaussian coordinate noise, rotation, and synthetic occlusion. The purpose is to measure how sensing degradation propagates through semantic segmentation, full-cloud reconstruction, structural components, and geometric traits. The report should therefore evolve from a single clean-data benchmark into sensitivity curves such as mIoU versus density, structural discrepancy versus dropout, and trait drift versus perturbation magnitude.

15.3 External and sensor-native data

Cross-dataset evaluation is required before making broad generalization claims. Pullally is a planned sensor-native integration target rather than a direct leaf/stalk benchmark. Its public release contains raw LiDAR and GNSS-RTK measurements from a quadruped robot in a Chilean experimental orchard, including Velodyne VLP-16 data and ROS bag files [4]. This makes it suitable for testing ingestion, coordinate transformation, registration, field-scale mapping, and the boundary between curated plant clouds and real robotic sensing. Additional annotated public 3D plant datasets can be considered separately for semantic-domain transfer.

15.4 Industrial interface

The repository already exposes reproducible CLI entry points and structured JSON/CSV/PLY artifacts, but a broader industrial interface remains planned. This stage should formalize stable request/response schemas, batch processing, automated technical reports, API/service boundaries, deployment packaging, model and configuration provenance, failure handling, performance targets, and operational monitoring.

The goal is to convert the research pipeline into a system that can be integrated without weakening the scientific traceability established in the current work.

15.5 Biological validation

Independent verification of MaizeField3D instance palette semantics and quantitative trait references remains important. Once reliable references are available, the current structural diagnostics can be extended to matched-instance IoU, leaf-count error, and trait MAE/RMSE. This should be performed as a separate validation layer rather than retroactively treating the present geometric proxies as biological ground truth.

16. Conclusion

Agri3D has progressed from a classical geometry baseline to a reproducible full-resolution 3D crop-structure analysis pipeline. The frozen XYZ-only PointNet++ model achieved approximately 0.8366 validation mIoU, outperforming the classical baseline and three subsequent controlled PointNet++ modifications. Deterministic nearest-neighbor projection preserved semantic quality at 100k-point resolution, reaching 0.8397 aggregate full-cloud mIoU across 78 validation plants.

The complete frozen Phase 4/5 pipeline executed successfully for all 78 plants in oracle and predicted semantic modes with zero failures. Structural extraction is deterministic and operational, but predicted semantics produce more components than oracle semantics on most plants, revealing downstream fragmentation. The current geometric phenotype outputs are reproducible and auditable, while biological accuracy remains intentionally unclaimed. Runtime analysis shows that instance graph processing, not neural inference, is now the principal end-to-end bottleneck.

The present system therefore provides a stable baseline for the next research stage. Future work will evaluate stronger 3D semantic architectures, controlled robustness perturbations, external and sensor-native datasets, verified biological references, and an industrial interface. The key methodological change is that future model improvements can now be judged by their effect on the full crop-structure pipeline rather than by segmentation mIoU alone.

References

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[3] Zhou, Q.-Y., Park, J., and Koltun, V. “Open3D: A Modern Library for 3D Data Processing.” 2018. https://arxiv.org/abs/1801.09847

[4] RAL-UC. “Pullally Dataset: LiDAR scans and GNSS measurements captured at the Pullally experimental fruit orchard, Chile.” Public dataset repository. https://github.com/RAL-UC/Pullally_Dataset

[5] Agri3D Crop Phenotyping repository. https://github.com/GuilhermeFrAlmeida/agri3d-crop-phenotyping

Appendix A. Reproducibility and Provenance

A.1 Core identities

Table A1. Reproducibility identifiers.

Item	Identifier / value
Frozen manifest SHA256	ae4446f5f832794d2464807ce82957190bd5178af4ab4a2b54c78972bc434371
Run 1 checkpoint SHA256	608a8f15319b0fc17daeefed92f1c278fd638a82adf30768c1c6d05a21ef03a9
Run 1 selected epoch	76
Frozen val78 code commit	3a50d684c944682b5f5c80054f3c034315824296
Model points	10,000
Source points	100,000 per segmented plant
Seed	42
Batch size	2
Frozen test split	78 plants; untouched

A.2 Frozen downstream configuration

Table A2. Frozen Phase 4/5 configuration used for val78.

Parameter	Frozen value
Coordinate units	meters
Vertical axis	Z
Projection	nearest_neighbor
Instance method	geometric_graph_v1
Neighborhood radius	0.020 m
Maximum neighbors	32
Minimum component size	100 points
Attachment diagnostic radius	0.03 m
Voxel volume size	0.01 m
Geodesic radius	0.03 m
Geodesic neighbors	8
Geodesic max nodes	2048

A.3 Artifact schema

Per-plant outputs include versioned phenotype JSON, semantic and instance label arrays, scalar-labeled metric PLY, sampled predictions for predicted mode, timing and provenance JSON, and checksum inventories. Batch outputs include per-plant and phenotype CSV tables, timing summaries, failure records, configuration, provenance, and an aggregate summary. The val78 run recorded zero failures.

A.4 Evidence-preservation policy

The validation artifacts are treated as immutable evidence. The frozen Phase 4A graph parameters were not changed after the 78-plant result was observed. Any future structural algorithm revision should be treated as a new experiment rather than a retroactive retuning of the same validation run.

Appendix B. Core Metrics

Table B1. Core current-stage metrics.

Metric	Value
Geometry baseline mIoU	0.622986
Run 1 sampled mIoU	0.836619
Run 1 projected full-cloud mIoU	0.839665
Run 2 mIoU	0.829574
Run 3 mIoU	0.831723
Run 4 mIoU	0.817336
Oracle mean structural components	15.936
Predicted mean structural components	17.923
Mean height	1.848 m
Predicted-mode mean total time	9.693 s/plant
Predicted-mode mean instance time	7.670 s/plant
Successful val78 plant-mode executions	156 / 156